

A Context-Grounded Question-Answering Assistant with Confidence-Based Abstention for Classroom Use

Huy Hoang Le
The Dewey Schools Tay Ho
Hanoi, Vietnam

Abstract—In this report I describe and evaluate a small question-answering (QA) assistant I built that answers a student’s question using *only* the notes a teacher provides, and that openly says it does not know when those notes do not support a confident answer. I started from a frustration with ordinary chatbots: they can give a fluent, confident answer that is simply wrong, and in a classroom, where the facts are the whole point, that is worse than no answer at all. My system avoids this by construction. It uses an extractive transformer model (DistilBERT) to copy an answer span straight out of the teacher’s notes, and it uses a single confidence threshold to decide when to answer and when to hold back. The focus of this report is measuring how well it works. I tested it on a knowledge base of three short documents on different subjects, using twenty questions, sixteen answerable from the documents and four deliberately outside all of them. It scored 18/20 for any threshold in $\tau \in [0.10, 0.30]$, correctly refused all four out-of-scope questions (which scored 0.08 or below, well separated from the answerable ones), and its only two mistakes were confidently *wrong* extractions rather than thresholding failures, a limitation I analyse as the natural ceiling of this simple approach.

Index Terms—question answering, extractive QA, DistilBERT, abstention, grounded answering, educational technology, natural language processing

I. INTRODUCTION

Students, myself included, increasingly reach for chatbots when they want a quick answer. But even the best general-purpose ones share a well-known weakness: they can sound confident and fluent while being completely wrong, a behaviour usually called hallucination. In a classroom this is a real problem. A tool that makes up a plausible-but-wrong date or name is arguably worse than one that stays quiet, because the student has no easy way to tell a good answer from a bad one. So I set out to build a small assistant that removes that risk by design. Instead of writing free-form text, it answers only by *extracting* a span of text out of documents the teacher provides, and it refuses when it is not confident the answer is really there. That leaves it just two possible responses to any question: a word-for-word quote from the trusted material, or

I am a high school student at The Dewey Schools Tay Ho (Adventure International Program). I designed, implemented, and evaluated this system myself. This work began as a self-directed project in grades 9–10 and was revised and evaluated for this report under fundamental concepts and feedback of Prof. Phan Duc Tri, National University of Singapore (NUS).

an honest “I could not find that in the notes.” It cannot state a fact that is not in the source.

In this report I focus on evaluating that tool. I explain how I built it and, mainly, how accurately it answers across a small multi-topic knowledge base, and how the single confidence threshold controls the balance between answering and staying quiet. I did not train the language model myself; I use an existing pre-trained one as a building block, so the part I claim as my own is the system around it, the grounding-and-abstention idea, and the evaluation.

II. BACKGROUND

A. Extractive Question Answering

Extractive QA is the task of taking a question and a passage of context and marking the shortest span of that passage which answers the question [2]. Unlike a generative model, an extractive one never writes new text; whatever it outputs is always a substring of the context it was given. I lean on exactly this property: because every answer my system produces is copied out of the teacher’s material, it is grounded by definition.

B. The DistilBERT Reading Model

For the reading model I chose DistilBERT [1], a compact, distilled version of BERT [3] that keeps most of BERT’s accuracy while being about 40% smaller (roughly 60% of BERT’s size) and is small enough to run comfortably on a CPU, which mattered to me because I wanted it to work on an ordinary laptop. The checkpoint I use, `distilbert-base-uncased-distilled-squad`, has already been fine-tuned on the Stanford Question Answering Dataset (SQuAD). Given a (question, context) pair it produces two sets of scores over the context tokens, one rating how likely each token is to *start* the answer and one for how likely it is to *end* it, and I turn those into an answer span and a confidence (Section III-C).

C. Grounding and Abstention

Two ideas are what make the system trustworthy. *Grounding* means an answer is tied to a specific source rather than to whatever the model happens to remember; extractive QA gives me this for free. *Abstention* means the system is allowed to say

no. A model that is forced to answer will, for an out-of-scope question, still hand back its best (low-confidence) guess, so I threshold the confidence and turn those weak guesses into honest refusals instead.

D. Relevance to Classroom Use

Because the model is small and runs locally, answering needs no paid API and no internet connection, which matters for a school that cannot count on either.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. Pipeline Overview

My assistant is a short Python program with a simple pipeline. At start-up it reads every plain-text file in a documents folder and joins them into a single context string. Then, for each question, the DistilBERT reader scores answer spans over that context, and I compare the confidence of the best span against a threshold to decide whether to return it or to abstain.

B. Document Ingestion and Context Assembly

I read all the .txt files in the documents folder and concatenate them into one string. This is what lets me grow the knowledge base: to add a subject I simply drop another file in the folder. For this report I put in three short documents on different subjects (Vietnamese history, the Solar System, and photosynthesis), and the assistant answered across all three with no change to the code.

C. Span Scoring

Let the reader produce start logits s_i and end logits e_j over the context token positions, where the logits are the raw scores the model outputs before they are turned into probabilities. I turn each set of logits into a probability distribution with the softmax function, which rescales a set of scores so they are all positive and sum to one, and I score the span running from token i to token j as the product of its two endpoint probabilities,

$$\text{score}(i, j) = \frac{\exp(s_i)}{\sum_k \exp(s_k)} \cdot \frac{\exp(e_j)}{\sum_k \exp(e_k)}. \quad (1)$$

The predicted span (i^*, j^*) is the one that maximises (1) subject to $i \leq j$ and a maximum answer length, and I take its confidence c to be the score the software (the Hugging Face transformers pipeline [4]) reports for that span: high (near 1) when the model is sure of both endpoints, and near 0 when it is not.

D. The Abstention Rule

I only let the system return the predicted span when its confidence clears a fixed threshold τ :

$$\text{output} = \begin{cases} \text{answer span } (i^*, j^*), & c \geq \tau, \\ \text{"I could not find that,"} & c < \tau. \end{cases} \quad (2)$$

That single parameter τ controls the entire trade-off between answering and refusing, so it is the main thing I study next.

score	verdict	answer	question
1.029	CORRECT	Hung Kings	Who ruled the Kingdom of Van Lang?
0.993	CORRECT	1975	When did the Vietnam War end?
0.831	CORRECT	Ngo Quyen	Who led Vietnam to independence at the Battle of Bach Dang?
0.972	CORRECT	1070	When was the Temple of Literature established?
0.658	WRONG	Hung Kings	Which dynasty repelled the Mongol invasions?
0.986	CORRECT	1945	When did French colonial rule in Vietnam end?
0.814	CORRECT	Jupiter	What is the largest planet in the Solar System?
0.915	CORRECT	Mercury	Which planet is closest to the Sun?
0.654	CORRECT	the iron oxide that covers its surface	Why is Mars called the Red Planet?
0.526	CORRECT	4.6 billion years ago	How old is the Solar System?
0.629	CORRECT	eight	How many planets are in the Solar System?
0.917	CORRECT	its bright system of rings	What is Saturn known for?
0.996	CORRECT	chlorophyll	Which pigment absorbs light during photosynthesis?
0.917	WRONG	carbon dioxide	What gas do plants release during photosynthesis?
0.302	CORRECT	the leaves	Where does photosynthesis mainly take place?
0.963	CORRECT	glucose	What sugar do plants produce during photosynthesis?
0.083	CORRECT	(refused)	What is the capital of France?
0.000	CORRECT	(refused)	Who won the 2018 World Cup?
0.000	CORRECT	(refused)	Who wrote the play Romeo and Juliet?
0.001	CORRECT	(refused)	What is the boiling point of water?

Score at threshold 0.15: 18/20 correct (16 answered, 4 refused)

Threshold sweep:
threshold=0.02 -> 17/20 correct
threshold=0.05 -> 17/20 correct
threshold=0.10 -> 18/20 correct
threshold=0.15 -> 18/20 correct
threshold=0.20 -> 18/20 correct
threshold=0.25 -> 18/20 correct
threshold=0.30 -> 18/20 correct
threshold=0.40 -> 17/20 correct
threshold=0.60 -> 16/20 correct

Saved chart to confidence_chart.png

Fig. 1. Sample per-question output from `confidence_scores.py` in the VS Code terminal, showing the extracted answer and reported confidence for each question.

IV. RESULTS AND EVALUATION

A. Experimental Setup

I evaluated the system with the pre-trained `distilbert-base-uncased-distilled-squad` reader on my CPU, over the three short documents described above. My test set is twenty questions: sixteen that are answerable from the documents (six on Vietnamese history, six on the Solar System, and four on photosynthesis) and four out-of-scope questions, such as “What is the capital of France?” and “Who won the 2018 World Cup?”, that I deliberately kept outside every document, so the correct behaviour there is to abstain. Both the documents and the questions are my own, and the set is small, so the numbers below describe how the system behaves rather than a benchmark score. For each question I record the confidence c the model reports, the answer it extracts, and whether that is the right outcome at the abstention threshold $\tau = 0.15$; Fig. 1 shows a sample of this raw output. The code and test set are available online.¹

B. Overall Accuracy

Table I breaks the result down by topic. At $\tau = 0.15$ the system answers 18 of the 20 questions correctly. Every one of the four out-of-scope questions is correctly refused: they receive confidences of 0.083 or below, far under the lowest confidence of any correct answer (0.302), so the confidence gap that the abstention rule depends on is wide and clean (Fig. 2). The two mistakes both occur on questions the documents *do* cover, and I analyse them below.

¹<https://github.com/Simbashrek276/Context-Grounded-AI-Chatbot->

TABLE I
ACCURACY BY TOPIC AT THRESHOLD $\tau = 0.15$.

Topic	Questions	Correct
Vietnamese history	6	5
Solar System	6	6
Photosynthesis	4	3
Out-of-scope (should refuse)	4	4
Total	20	18

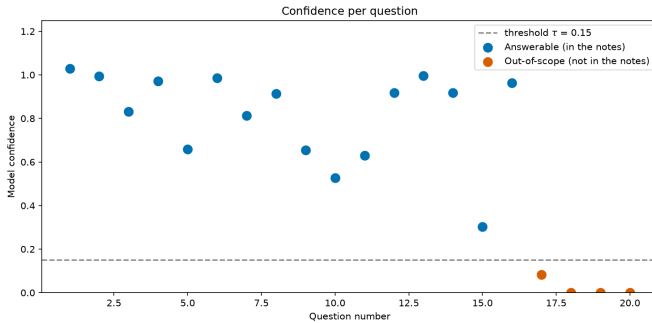


Fig. 2. Reported confidence for each question. Answerable questions sit well above the out-of-scope ones, and any threshold in the shaded band separates the two groups cleanly.

C. Threshold Sweep

To study the abstention rule I swept the threshold τ and measured overall accuracy each time, counting an answerable question right only if answered correctly and an out-of-scope question right only if refused. Table II shows that accuracy holds at its best value, 18/20, for the whole band $\tau \in [0.10, 0.30]$. Below it, an out-of-scope question slips over the bar and is wrongly answered; above it, the lowest-confidence correct answers start getting refused. The wide flat band means the threshold does not need careful tuning.

D. Discussion and Error Analysis

The evaluation makes two points that I think matter more than the headline score. First, abstention works exactly as intended for questions the notes do not cover. All four out-of-scope questions scored at or below 0.083, while every correct answer scored at least 0.302, so any threshold in the $[0.10, 0.30]$ band separates them perfectly. The system reliably refuses what it should not answer. Second, and this is the honest limit, the two errors are not thresholding problems but confident misreadings. Asked “Which dynasty repelled the Mongol invasions?” the model answered “Hung Kings” (confidence 0.658) instead of the Tran dynasty, and asked “What gas do plants release?” it answered “carbon dioxide”

(confidence 0.917), which is the gas plants take *in*, not the one they release (oxygen). Because both wrong answers are given with high confidence, no threshold can filter them out. The confidence here is a product of two span probabilities, not a calibrated measure of whether the answer is correct, which is exactly why a confident answer can still be wrong. This is the ceiling of the approach: abstention protects against questions outside the material, but it cannot protect against the reader confidently misreading a passage that *is* in the material. Recognising that limit is, to me, the most useful outcome of the evaluation.

V. CONCLUSION

In this report I described and evaluated a small, fully local question-answering assistant that answers strictly from teacher-provided material and backs off when it is not sure. Using an extractive DistilBERT reader and a single confidence threshold, it gives grounded answers by construction and turns low-confidence guesses into honest refusals. On a twenty-question test set spanning three subjects it reached 18/20 for any threshold in $\tau \in [0.10, 0.30]$, cleanly refused all four out-of-scope questions, and made only two mistakes, both confident misreadings rather than thresholding failures. I kept the design deliberately minimal (plain-text input, one reading model, one threshold) so that I could actually understand it, run it without paying for anything, and point it at a new subject just by adding a file.

ACKNOWLEDGMENT

I designed, implemented, and evaluated this system myself, as a self-directed project that began in grades 9–10. I am grateful to Prof. Phan Duc Tri of the National University of Singapore (NUS), who introduced me to the foundational concepts behind the work and gave feedback on this report. The reading model and dataset are the work of their respective authors, cited below, and are used here as building blocks.

REFERENCES

- [1] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” in *Proc. 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing (NeurIPS)*, 2019.
- [2] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, “SQuAD: 100,000+ questions for machine comprehension of text,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2016, pp. 2383–2392.
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [4] T. Wolf *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proc. EMNLP: System Demonstrations*, 2020, pp. 38–45.

TABLE II
ACCURACY OVER THE TWENTY-ITEM TEST SET VERSUS THRESHOLD τ .

τ	0.05	0.10	0.15	0.20	0.25	0.30	0.40	0.60
Correct /20	17	18	18	18	18	18	17	16